

1. GENERAL WELL INFORMATION

Well Name	Alpha-01	Well ID	W001
Field	Field_Alpha	Block	Block_1
Spud Date	02-05-2024	Rig Release Date	29-05-2024
Total Depth (TD)	3050 m	Rig Name	Rig Sagar-3
Well Type	Vertical Exploratory	Trajectory	Vertical

2. FORMATION INFORMATION

Formation_X (unconsolidated sand-shale sequence) was encountered between **2850 m** and **3100 m**. This interval is historically associated with high permeability streaks and known losses of circulation in offset wells within Field_Alpha.

3. LITHOLOGY SUMMARY

Interval (m)	Formation	Lithology	Notes
2500-2850	Formation_W	Shale, grey, firm	Normal drilling trend
2850-3020	Formation_X	Sand, friable, interbedded shale	Loss-prone; monitor returns
3020-3050	Formation_Y	Shale, grey-black, hard	Tight hole tendency below 3140 m

4. HISTORICAL DRILLING EVENTS

Event 1: Partial Mud Loss at approximately 2905 m

Observation: Reduction in mud returns was observed while drilling through the friable sand streak of Formation_X.

Engineering Action: Drilling was temporarily stopped. Loss Circulation Material (LCM) was pumped and mud weight was reduced from 1.22 SG to 1.18 SG.

Outcome: Mud losses decreased from approximately 8 bbl/hr to negligible levels. Drilling resumed without further incident.

Event 2: Stuck Pipe (Differential Sticking) at approximately 3150 m

Observation: Overpull of 35 klb was noted while making a connection; string would not rotate or reciprocate freely.

Engineering Action: Circulation was maintained, spotted a pipe-freeing pill (diesel/oil based), and worked the string with jarring at reduced RPM.

Outcome: String was freed after approximately 6 hours of remedial work. Drilling resumed with mud weight held at 1.24 SG through the interval.

5. CASING AND COMPLETION SUMMARY

Casing	Size (in)	Set Depth (m)	Cement Top (m)
Conductor	20	60	Surface

Surface	13-3/8	650	Surface
Intermediate	9-5/8	2450	1200
Production	7	3050	2200

6. RECOMMENDATION

Monitor mud returns closely while drilling through Formation_X between 2850 m and 3020 m; keep LCM pills pre-mixed and ready above 2870 m. In the interval below 3120 m, minimize static time and maintain adequate mud weight to reduce differential sticking risk.

This Well Completion Report is a synthetically generated historical record produced for the NWIS prototype (W001). Not associated with an actual drilling operation.